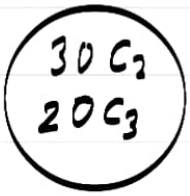
Machine Learning

24-04-26

Decision Tree Classifier:



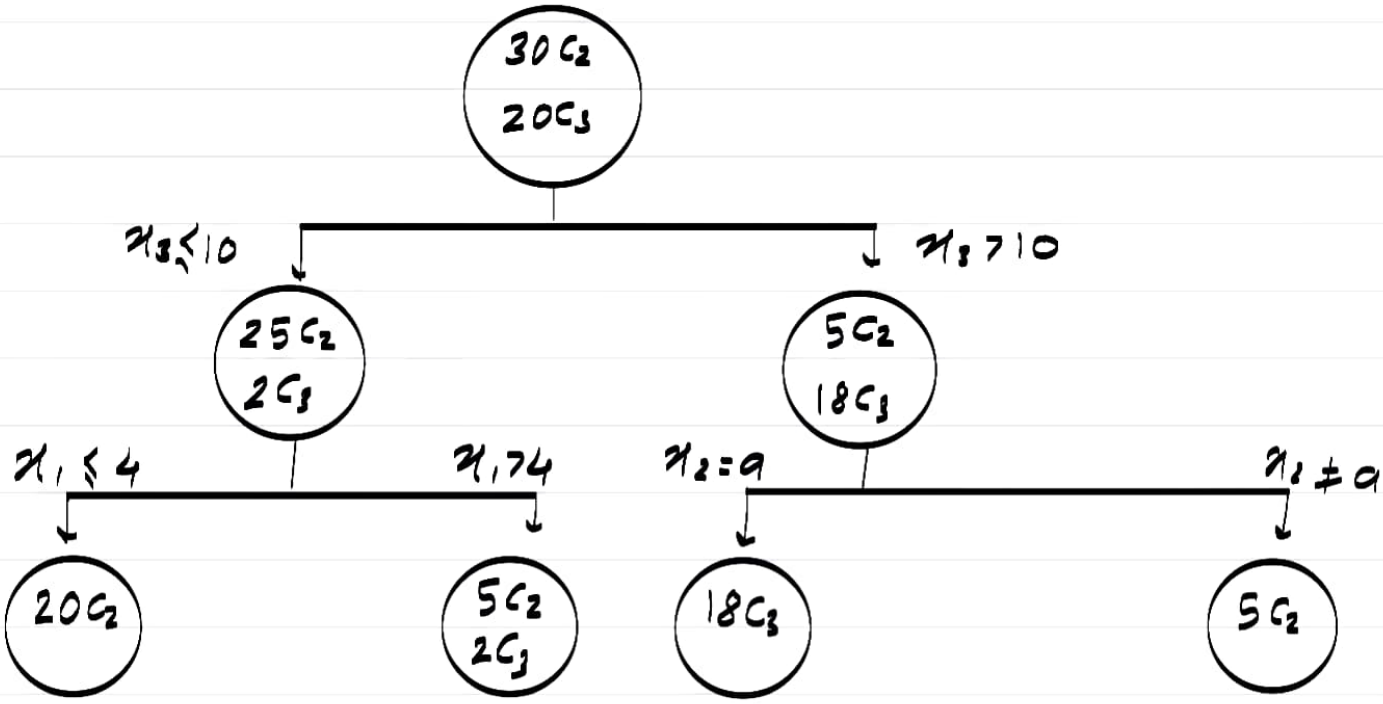
- randomly pick 2 features and formulate decision rules

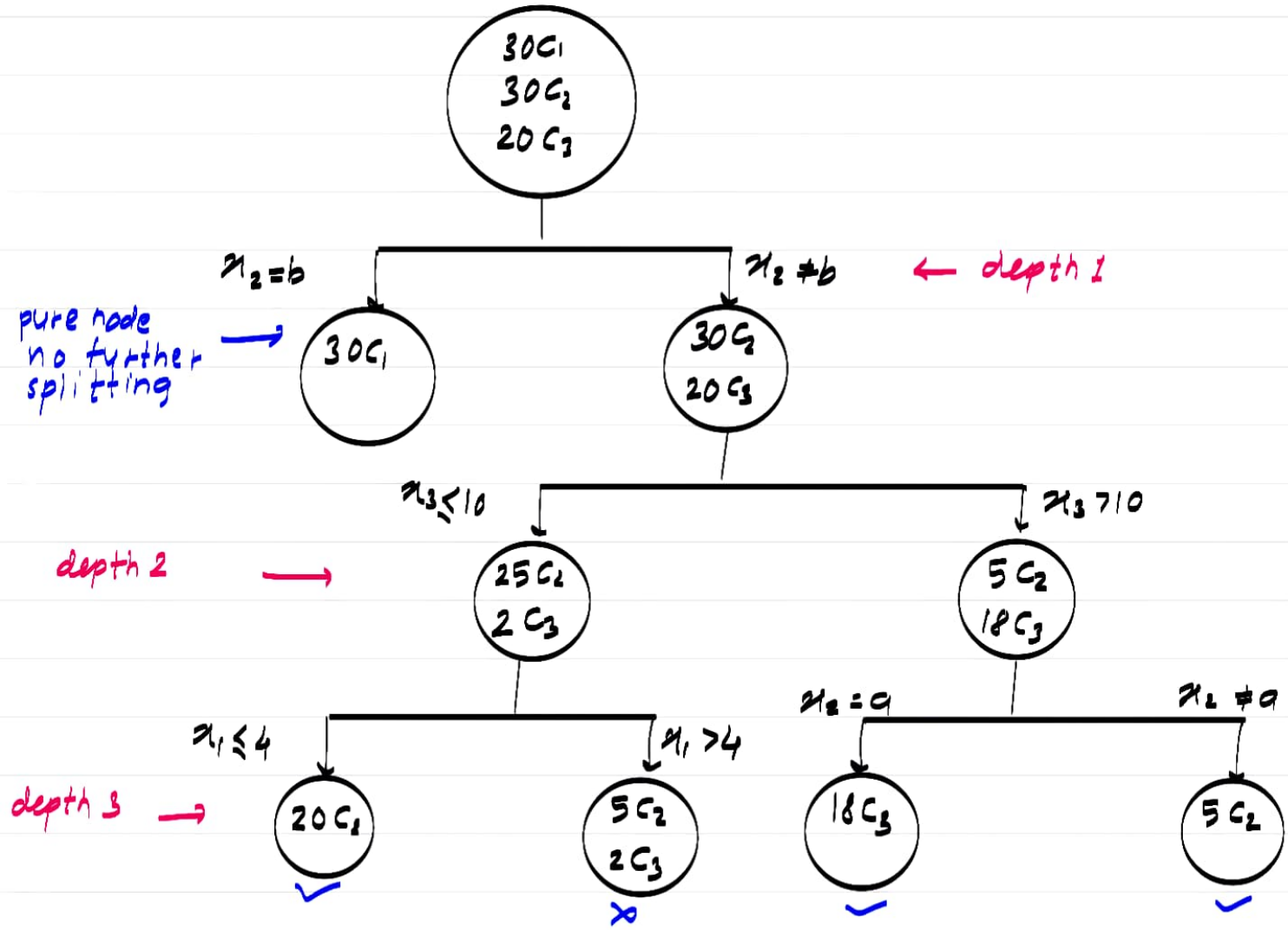
  lets say we picked x_2 and x_3

  x_2 → categorical → 3 categories rules

  x_3 → continuous → 100 percentiles rules

- lets say best decision rule found is $x_3 \leq 10$





The training of DT model stops when max depth is reached.

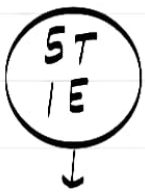
lets say we decided max-depth = 3 then DT training will stop at depth 3

The splitting of node stops when

1) node has become 100% pure

2) node has less datapoints than 'min samples split'

eg.



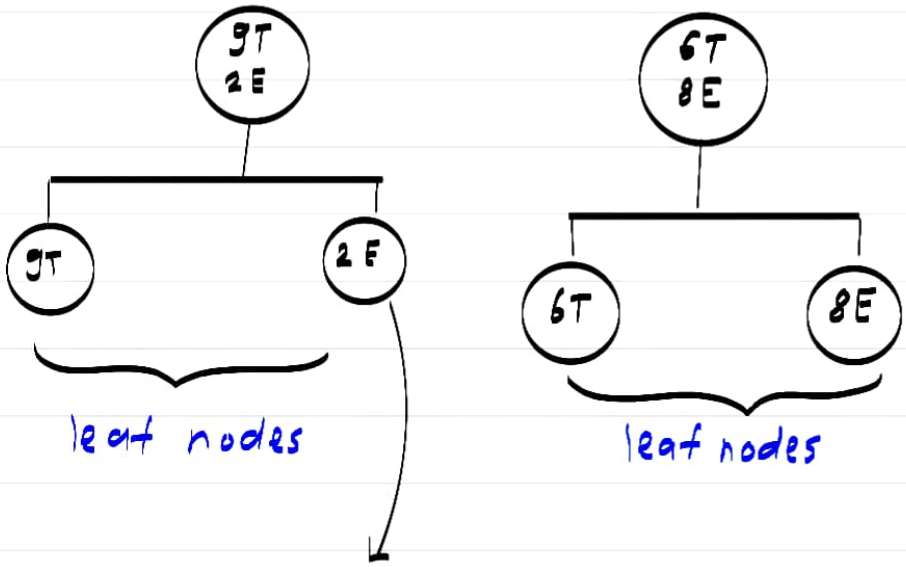
as 6 < 9 this node won't be splitted further

as 10 > 9 this node will be further splitted

lets say min samples split = 9

- split of a parent node stops if its child has number of datapoints less than 'min samples leaf'

eg.



lets say
min samples split = 10
min samples leaf = 4

n (child node) = 2 which is less than 4 (min samples leaf)
due to this the parent won't go for split and this split would be reverted back.

max depth : stops DT training when max depth is reached

max features: decides number of features to be used at every split to find best decision rule

min samples split: decides whether further splitting of parent node is needed or not

min samples leaf: checks if child/leaf nodes have enough datapoints or not after the split.

entropy : $-\sum_{i=1}^{k} p_i \log(p_i)$

- log operation in this equation is comparitively slow
- at every level we need more than 100 decision rules to test and the log operation becomes bottleneck for it
- we can use a faster alternative for entropy that is <u>Gini Index</u>

Gini Index

- computes impurity of node

GI $\propto$ impurity

$$GI = 1 - \sum_{i=1}^{k} (p_i)^2$$

k - no. of classes in given node
p_i - proportion of ith class in given node

3T
5E

$$GI = 1 - \sum_{i=1}^{2} p_i^2$$

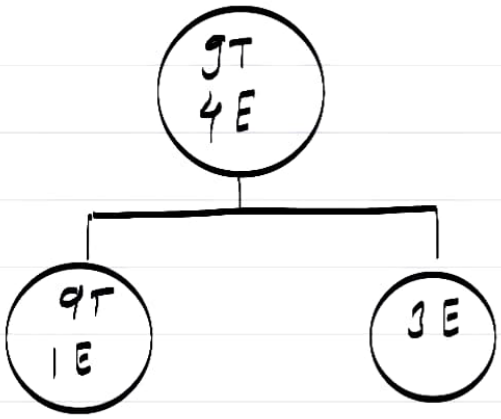
$$= 1 - \left[\left(\frac{3}{8}\right)^2 + \left(\frac{5}{8}\right)^2 \right]$$

$$= 1 - \left[\frac{9+25}{64} \right] = \frac{30}{64}$$

$$GI = \frac{30}{6}$$

information gain = entropy before split − entropy after split

information gain = gini index before split − gini index after split



$$GI\ parent = 1 - \left[\left(\frac{9}{13}\right)^2 + \left(\frac{4}{13}\right)^2 \right]$$

$$= 1 - \left[\frac{81+16}{169} \right] = \frac{72}{169}$$

$$GI\ child1 = 1 - \left[\left(\frac{9}{10}\right)^2 + \left(\frac{1}{10}\right)^2 \right]$$

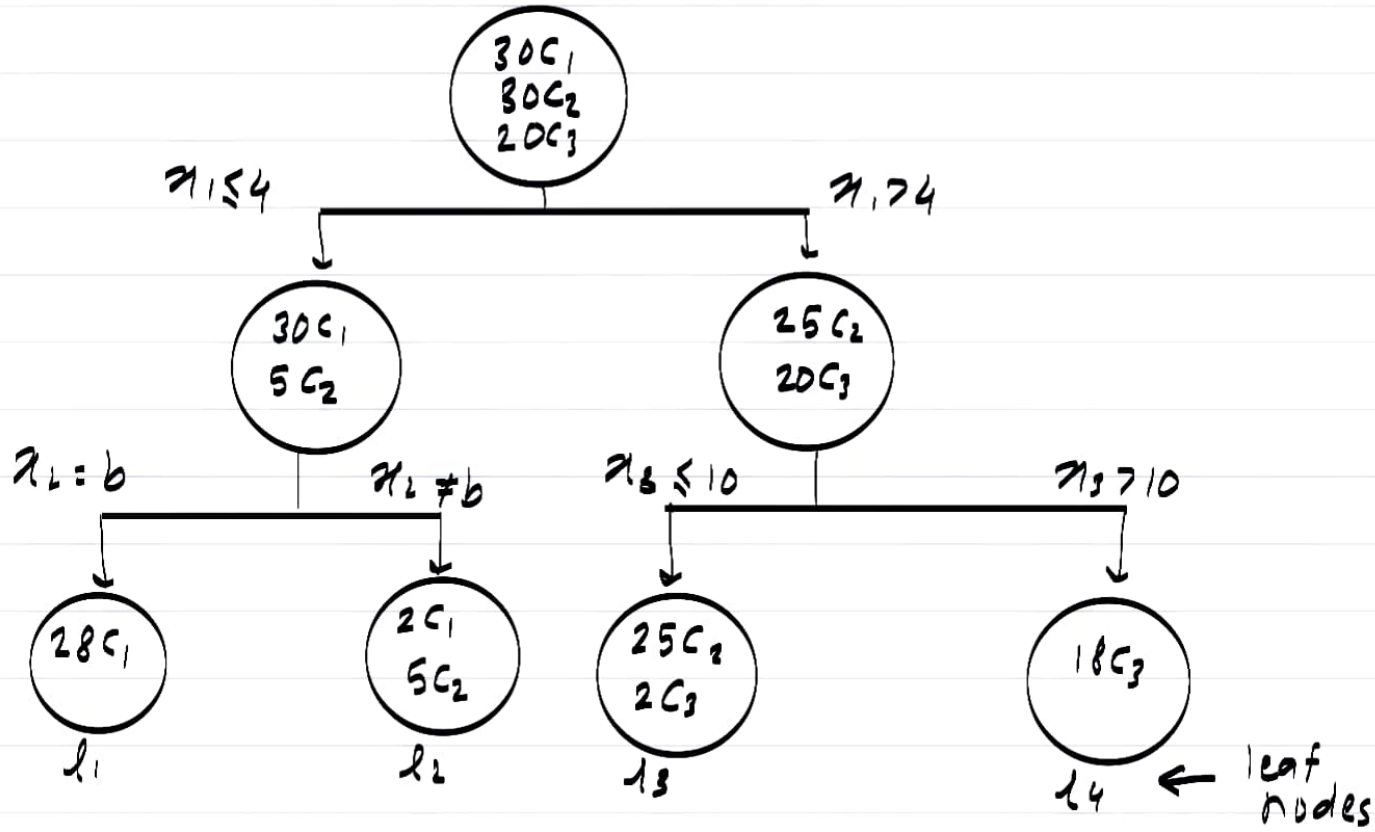
$$= 1 - \left[\frac{81+1}{100} \right] = \frac{18}{100}$$

$$GI\ child\ 2 = 1 - \left[\left(\frac{3}{3}\right)^2 \right] = 1 - 1 = 0$$

$$IG = GI\ parent - \left[\frac{n(child1)}{n(parent)} \times GI\ child1 + \frac{n(child2)}{n(parent)} \times GI\ child2 \right]$$

$$= \frac{72}{169} - \left[\frac{10}{13} \times \frac{18}{100} + \frac{3}{13} \times 0 \right] = 72/169 - 18/130$$

Prediction for new datapoints



$x_1 = 8$
$x_2 = c$
$x_3 = 7$

→ belongs to l_3 leaf node

the class in majority in l_3 is C_2

hence final model prediction = C_2

final output of model is found using majority voting in leaf node.